

Week-4 (Practice Problems)
Strings

(1)

Write a java code that will take user input as a string and the output should tell whether the input is palindrome or not.

Input	Output
hello Explanation: If we reverse the word hello then we get olleh, now hello is not equal to olleh so it is not palindrome.	hello is not palindrome
madam Explanation: If we reverse the word madam then we get madam, now madam is equal to madam so it is palindrome.	madam is palindrome

(2)

Write a java code that will take one input from the user with numbers separated by comma, your task is add all the prime numbers from that string and display the summation.

Input	Output
2,3,11,6,100,4	16
4,9,122,1000,42	0

(3)

Zark Muckerberg is worried about his security for the messaging system. He wants to ensure that all messages are encrypted before it is sent. Now, Zark Muckerberg hired you to write a code that will encrypt the message. To encrypt first you need to count how many times each letter exists and then change the letter to that many times. For example. "Hello" is the message here H, e, l and o exists 1,1,2 and 1 times respectively. Now, change H to l as H exist only 1 time, e is changed to f as e exists only 1 time, then l is changed to n as l exists 2 times. Thus the final message is "lfnnp".

Input	Output
Hello	lfnnp
zzzxy	ccczzz

(4)

Write a python code that will tell you how many vowels and consonants are there in a single word given by the user.

Input	Output
hello	Vowels: 2 Consonants: 3
ROnaIDo	Vowels: 3 Consonants: 4

(5)

After completing Task:3 Zark Muckerberg was very happy with your work, now he wants to make sure that users will be able to know whether their password is strong or not. He figured out some requirements for a strong password which are: **length ≥ 6** , contains **uppercase**, **lowercase**, **symbols** and **numbers**. Your task is to write a code that will make sure all these requirements are followed, if something is missing then it should what are missing.

Input	Output
password	Uppercase missing Symbol missing Number missing
pass12	Length should be greater than 6 Uppercase missing Symbol missing
P@ssWord123	Strong Password

(6)

Write a code that will take an input from the user and convert all uppercase characters to lowercase and all lowercase characters to uppercase, finally showing the result.

Input	Output
RonALdO	rONaIdO
FreDERiCo CHiesA	fREderlC O chIESa

(7)

First user will enter the name of a football player and then enter the name of a cricket player, then you have to merge the first name of the football player with the last name of the cricket player and show the result.

Input	Output
Christiano Ronaldo Shakib Al Hasan	Christiano Hasan
Frederico Chiesa Mitchell Starc	Frederico Starc

(8)

You are hired to write a program for an elementary school that will make corrections to sentences written by students. Requirements are as follows: first letter will always be capital letter, i will be always uppercase if single and after full stop, exclamation and question mark the next letter should be capital.

Input	Output
i am a football player. the name of my club is Real Madrid.	I am a football player. The name of my club is Real Madrid.
what is your name? i need to know some information about you before you are allowed to register. are you ok with it?	What is your name? I need to know some information about you before you are allowed to register. Are you ok with it?
Messi is the goat. he plays for Argentina. his jersey number is 10. currently he plays for Inter Miami.	Messi is the goat. He plays for Argentina. His jersey number is 10. Currently he plays for Inter Miami.

(9)

You need to write a code in python to develop a calculator that performs addition and subtraction. In the calculator the user will enter a string with operator (+ or -) and you need to display the final result.

Input	Output
5+3+10+2-10	Result: 10
7+100-10+7+3-1	Result: 106
1000+2000-8000-100+5	Result: 5095